BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

BIOLOGY

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Biology (Paper - I) (Objective Type) (Group - I)

Time Allowed: 15 Min. | **Maximum Marks:** 12

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer -book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Which one ecologist developed the concept of ecological pyramids:

(A) Mendel
(B) Charles Elton
(C) Darwin
(D) Lamarck

Q2. In which group, external fertilization takes place:

(A) In fishes
(B) In reptiles
(C) In birds
(D) In mammals

Q3. Which one drug weakens the short term memory in human.

(A) Marijuana
(B) Morphine
(C) Codeine
(D) Psilocin

Q4. Which one of these is a muscular structure present below the lungs:

(A) Trachea
(B) Voice box
(C) Alveoil
(D) Diaphragm

Q5. Interferons were first time produced in:

(A) 1974
(B) 1976
(C) 1978
(D) 1980

Q6. The concave side of kidney which faces verteral column is called:

(A) Hilus
(B) Renal cortex
(C) Renal medulla
(D) Renal pelvis

Q7. Which one is the reservoir of free gaseous nitrogen:

(A) Producers
(B) Consumers
(C) Atmosphere
(D) Water

Q8. How many bones make our vertebral column:

(A) 20
(B) 22
(C) 24
(D) 26

Q9. Which one part of brain is present on the top of medulla:

(A) Mid brain
(B) Pons
(C) Cerebrum
(D) Thalamus

Q10. In which one histone protein is present.

(A) DNA
(B) RNA
(C) Chromosomes
(D) Centrosome

Q11. After fertilization in plants the seed develops from:

(A) Petals
(B) Sepals
(C) Ovule
(D) Ovary

Q12. Which one is source of variations:

(A) Guttation
(B) Transcription
(C) Translation
(D) Mutation

Biology (Paper - I) (Subjective Type) (Group - I)

Time Allowed: 1.45 Min. | **Maximum Marks:** 48

Q2. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Describe structure and function of alveolus.

(ii) Write syptoms of asthma.

(iii) Describe two harmful effects of smoking on human health.

(iv) What is meant by homeostasis?

(v) Define osmosis.

(vi) What is meant by receptors?

(vii) Briefly describe structure and function of motor neurons.

(viii) Write the functions of insulin and glucagon.

Q3. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Write two big functions of skeletal system in our body.

(ii) Differentiate between compact bone and spngy bone.

(iii) Define reproduction and name its two basic types.

(iv) What is meant by alternation of generation?

(v) Differentiate between gene and allele.

(vi) Write two main sources of variations.

(vii) What is meant by Punnett square?

(viii) Define Mendel's law of independent assortment.

Q4. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Differentiate between primary and secondary consumers.

(ii) Write down examples of carnivores plants.

(iii) What is green house effect?

(iv) Write symptoms of dengue fever.

(v) Define fermentation.

(vi) Who was Professor Scrimshaw?

(vii) What is pharmaceutical medicine?

(viii) Define vaccine.

Biology (Paper - I) (PART - II) (Group - I)

Time Allowed: 1.45 Min. | **Maximum Marks:** 48

> Note: Attempt any TWO questions.

Q5(a). Explain the function of kidney.

Q5(b). What is cartilage? Write about its three types.

Q6(a). Describe peripheral nervous system and write its types with their functions.

Q6(b). Explain process of budding in yeast with the help of diagram.

Q7(a). What is meant by biotechnology? Describe the application of biotechnology in the field of medicine.

Q7(b). Write the name of sources of drugs and explain any three.

GROUP – II

Biology (Paper - I) (Objective Type) (Group - II)

Time Allowed: 15 Min. | **Maximum Marks:** 12

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer -book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Which one is main producer in aquatic ecosystem:

(A) Fungi
(B) Bacteria
(C) Phytoplankton
(D) Protozoans

Q2. Part of testes provides nutrients ot sperms:

(A) Collecting ducts
(B) Seminal vesicles
(C) Prostate glands
(D) Cowper's glands

Q3. It effects on central nervous system and causes drowsiness:

(A) Morphine
(B) Mescaline
(C) Psilicon
(D) Codeine

Q4. During exercise or other physical works breathing rate per minute increases up to:

(A) 20-30 minutes
(B) 30-40 minutes
(C) 40-50 minutes
(D) 50-60 minutes

Q5. In which year, human genome project was launched:

(A) 1960
(B) 1970
(C) 1980
(D) 1990

Q6. Water lily is an example of:

(A) Hydrophytes
(B) Xerophytes
(C) Halophytes
(D) Mesophytes

Q7. The current level of urbanization in Pakistan is about:

(A) 22%
(B) 32%
(C) 42%
(D) 52%

Q8. Elastic cartilage is found in:

(A) Epiglottis (B) Larynx

(C) Bronchial tubes (D) Inter vertebral discs

Q9. It is the union of several axons:

(A) Dendrties (B) Nerve

(C) Schwann cells (D) Node of Ranvier

Q10. Movement of genes from one population to another is called:

(A) Crossing over (B) Recombination

(C) Gene flow (D) Mutations

Q11. Mint and chrysanthemum reproduce by:

(A) Bulbs (B) Corms

(C) Stem tubers (D) Suckers

Q12. Process of translation occurs in:

(A) Golgi apparatus (B) Ribosomes

(C) Nucleus (D) Endoplasmic reticulum

Biology (Paper - I) (Subjective Type) (Group - II)

Time Allowed: 1.45 Min. | **Maximum Marks:** 48

Q2. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Define cellular respiration.

(ii) Differentiate between inspiration and expiration.

(iii) How do pleural membranes protect lungs?

(iv) What is meant by thermoregulation?

(v) What is lithotripsy?

(vi) Distinguish between dendrites and axons.

(vii) Define reflex arc.

(viii) Write the cause and two symptoms of hyperthyroidism.

Q3. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Write difference between compact bone and spongy bone.

(ii) What is joint? Write its types.

(iii) Describe the process of parthenogenesis in honey bee.

(iv) Differentiate between regeneration and binary fission.

(v) Differentiate between homozygous genotype and heterozygous genotype.

(vi) Define monohybrid and dihybrid cross.

(vii) Differentiate between breeders and cultivars.

(viii) Write the sources of variations.

Q4. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Differentiate between population and community.

(ii) Define biosphere.

(iii) Differentiate between primary and secondary consumers.

(iv) What is meant by biological nitrogen fixation?

(v) Write the name of two enzymes used in genetic engineering.

(vi) What is fermenter?

(vii) What is the difference between pharmacology and pharmacy?

(viii) What is the use of vaccines? Write example.

Biology (Paper - I) (PART - II) (Group - II)

Time Allowed: 1.45 Min. | **Maximum Marks:** 48

> Note: Attempt any TWO questions.

Q5(a). What is the function unit of kidney? Describe its structure with labelled diagram.

Q5(b). Explain antagonism in muscle action with biceps and triceps as example.

Q6(a). Explain pituitary gland in detail.

Q6(b). Draw and label structure of a dicot seed and explain its different parts in detail.

Q7(a). Write down four achievements of genetic engineering.

Q7(b). Write mode of action of vaccine.